SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1714

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I V Murala Karthik (192312567) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	24
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	24
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	18
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	18
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	6

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CO3-AT1

1) a) Vehicle Fault Diagnosis System:-

Propositional Logic Symbols:-

- EWL = Engine Warning Light
- UN = Unusual Noise
- LOP = Low oil pressure
- OS = Overheating Sign
- EFR = Engine failure Risk
- OL = Oil Leak
- CSF = Coolant System failure

Rules:-

1, $EWL \land UN \Rightarrow EFR$

2, $EWL \land LOP \Rightarrow OL$

3, $UN \land OS \Rightarrow CSF$

b) Forward chaining.

Step	Rule	Condition.
1	1	$EWL \land UN$ = True $\land$ True
2	3	$UN \land OS$ = True $\land$ True
3	2	$EWL \land LOP$ = True $\land$ False

Derived faults:- Engine failure Risk, Coolant System failure.

c) Backward chaining - Goal: CSF = True.

Goal: CSF

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Rule 3 -> Subgoal 1 ; Subgoal 2 -> CSF verified.

2) Hospital Emergency Ward Routing (FOL)

a) Unification with Rule 1:-

Rule 1:- $\forall x$: Patient(x) $\land$ Critical Condition(x) $\Rightarrow$ priority Corridor(x)

Substitution (MGU):

- $\Theta_1 = \{x \mid \text{Patient A}\}$
- patient (patientA)
- Critical condition (Patient A)
- Derived:- priority Corridor (patient A).

patient B:- Cannot Unify

b) Resolution - Prove Hold (patient B):-

Step	clause 1	clause 2
1	Hold (PatientB)	Rule 3
2	Resolvent-1	Patient (patientB)
3	Resolvent-2	Waiting Behind (B,A)
4	Resolvent-3	Corridor Cleared (A)
Hold (PatientB) = TRUE		

c) Proposed Additional Rules:-

- $\forall x \forall y$:- Severity Level $(x, L_1) \land$ Severity Level $(y, L_2) \land L_1 > L_2 \Rightarrow$ Priority(x)
- $\forall x \forall y$: corridor Cleared(x) $\land x \neq y \Rightarrow$ Corridor Cleared(y)

Flight Scheduling Expert System:

Knowledge Base:

- Storm Warning => Flight Delay Needed
- Flight Delay Needed ∧ International Route => Notify passengers
- Notify passengers ∧ International Route => Compliance Risk.

Forward chaining:-

Step	Rule	Derivation
1	P_1	Storm warning => Flight delay Needed
2	P_2	Flight delay Needed ∧ International Route => Notify Passengers.
3	P_3	Notify passengers ∧ International Route => Cannot Fire.